Trainings

Preparatory Steps

⚠ WARNING ⚠

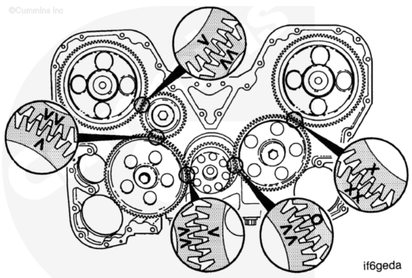
This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.

- Remove the front gear cover and all related components. Refer to Procedure 001-031 (Gear Cover, Front) in Section 1. (/qs3/pubsys2/xml/en/procedures/28/28-001-031-tr.html)

Remove

⚠ CAUTION ⚠

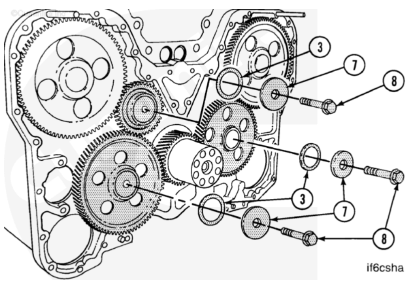
The three idler gears, the two camshaft gears and the crankshaft gear have index marks that are aligned at assembly. Improper alignment will cause extensive engine damage.



Mark the gears ("A") to make sure they are reassembled in the proper position. Mark the gears ("B") to show if the engine has been barred over while the camshafts were removed.

⚠ CAUTION ⚠

Do not allow the gears or shafts to fall or damage to the parts will result.



Note : It is recommended that the parts for each gear and shaft assembly be kept together to aid future

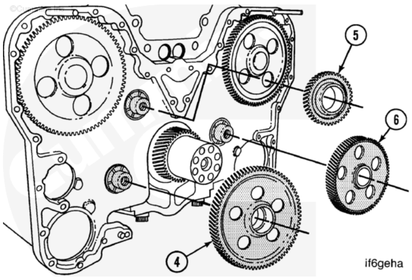
assembly.

Note : It is recommended that the gear lash and the end clearance be measured before removal to identify any worn gears or thrust bearings.

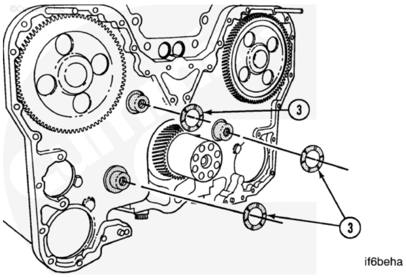
Remove the three capscrews (8). Remove the three retainers (7). Remove the three outer thrust bearings (3).

Remove:

4. Water pump idler gear
5. Right bank camshaft idler gear
6. Left bank camshaft idler gear.



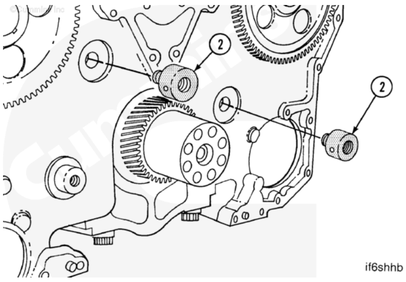
Remove the three inner thrust bearings (3).



Note : A slide hammer type tool that contains [3/4x16 inch) threads can be used to remove the idler shafts.

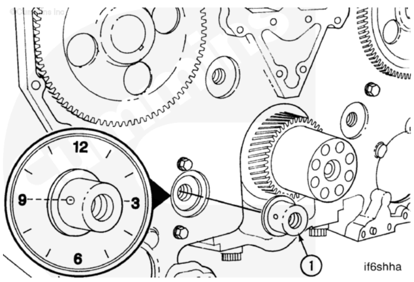
Remove the two camshaft idler gear shafts (2).

Carefully check the shafts and the bore in the cylinder block for any indication of movement from the shaft.



Check to see if the water pump idler shaft (1) has remained in alignment as shown (oil supply hole at 9 o'clock position). If there is any indication that the shaft has moved, be sure to check the parts thoroughly.

Remove the shaft.

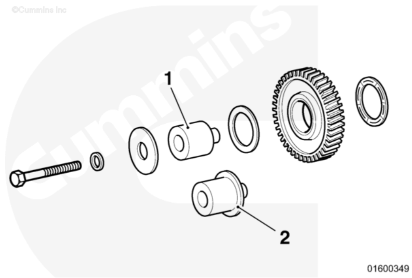


Inspect for Reuse

Fixed Time

⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.

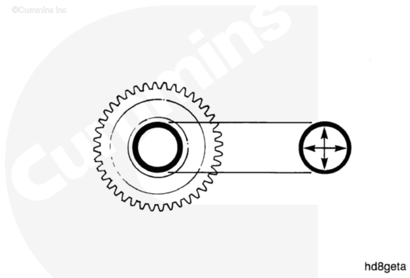


The shafts for camshaft idler gears and the water pump idler gears are different. The camshaft idler shafts (1) do **not** have an integral flange like the water pump idler shaft (2).

Clean and check the parts for damage.

Measure the inside diameter.

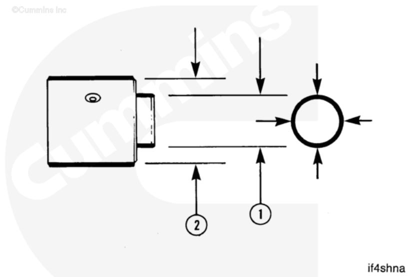
mm		in
47.638	MIN	1.8755
47.714	MAX	1.8785



The bushing in the gear is precision bored after installation. If machining capability is **not** available, replace the bushing and the gear as an assembly.

⚠ CAUTION ⚠

Mounting parts for the water pump Idler shaft is changed on newer engines. This flanged capscrew can be used if proper clearance and torque values are also applied.



Note : Some engines have a camshaft idler shaft that is the same as the water pump idler shaft.

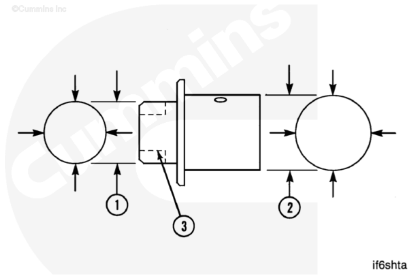
Measure the outside diameter

mm		in
25.397	MIN	0.9995
25.400	MAX	1.0000

mm		in
47.549	MIN	1.8720

mm		in
47.600	MAX	1.8740

Water Pump Idler Shaft outside diameter



mm		in
43.167	MIN	1.6995
43.180	MAX	1. 7000

mm		in
47.549	MIN	1.8720
47.600	MAX	1.8740

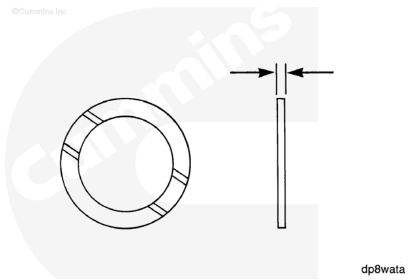
mm		in
25.397	MIN	0.9995
25.400	MAX	1.0000

mm		in
47.549	MIN	1.8720
47.600	MAX	1.8740

Note : Some engines with an engine serial number (ESN) lower than 33110701 have the shaft with the larger diameter (1). These engines were modified in the field until a service tool that is available to modify the cylinder block.

Check the grooved side for damage.

Measure the thickness.

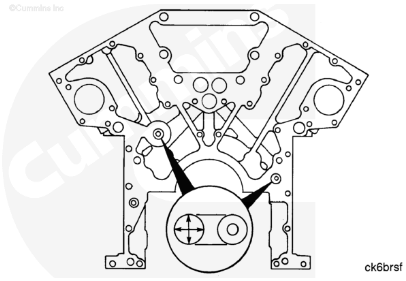


mm		in
2.34	MIN	0.092
2.39	MAX	0.094

Measure the inside diameter of the bore for the camshaft idler shafts.

Note : Some engines have camshaft idler shaft bores that have the same inside diameter specification as the

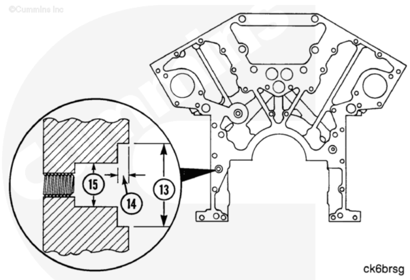
new water pump idler bore. These engines have engine serial numbers between 33110533 and 33110742. Refer to Procedure 001-026 (Cylinder Block) in Section 1. (/qs3/pubsys2/xml/en/procedures/28/28-001-026-tr.html)



mm		in
25.38	MIN	0.999
25.40	MAX	1.000

If the bore is **not** within specifications, a repair sleeve can be installed. See the Alternative Repair Manual, Bulletin 3379035.

The bore for the water pump idler shaft can be one of two designs.



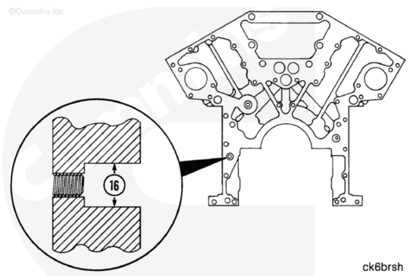
Newer cylinder blocks and blocks that have been repaired for the new shaft have a counterbore (13) and (14).

mm		in
43.18	MIN	1.700
43.22	MAX	1.701

mm		in
4.57	MIN	0.180
5.08	MAX	0.200

mm		in
21.97	MIN	0.865
22.23	MAX	0.875

Older cylinder blocks do **not** contain the counterbore.



mm		in
25.38	MIN	0.999
25.4	MAX	1.000

An idler bore kit, Part Number 3822548, is available to modify the bore for the new shaft. The old shaft is available for service.

Install

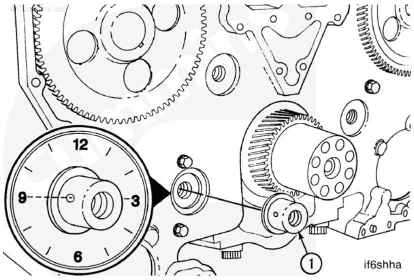
⚠ CAUTION ⚠

Newer and older engines do not contain the same parts. If the correct parts are not installed, the engine will be seriously damaged.

⚠ CAUTION ⚠

The water pump Idler shaft. (1) must be installed so that the all holes on the outside diameter of the shaft are at the 3 o'clock and 9 o'clock positions. If the all holes of the shaft are not In the correct position, failure will result due to lack of lubrication of the bushing. It Is recommended that the two camshaft idler shafts be installed with the holes in the same position as the water pump idler shaft.

Mounting parts for the water pump idler shaft have changed on newer engines. This flanged capscrew can be used if proper clearance and torque values are applied.

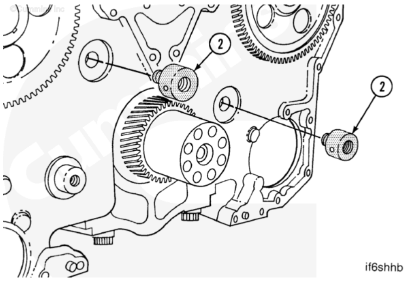


Install the water pump idler shaft (1).

Rotate the shaft until the oil holes are at the 3 o'clock and 9 o'clock positions as shown.

Install the two camshaft idler shafts (2).

Note : If a camshaft idler shaft does not stay in the block, check to be sure the parts are within specification.

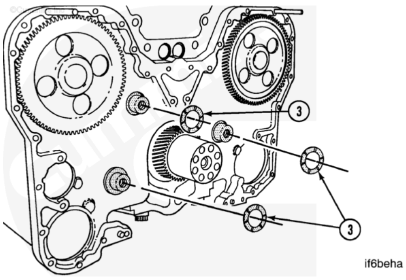


⚠ CAUTION ⚠

The grooves on all thrust bearings (3) must be toward the idler gears. If the parts are installed wrong, failure will result.

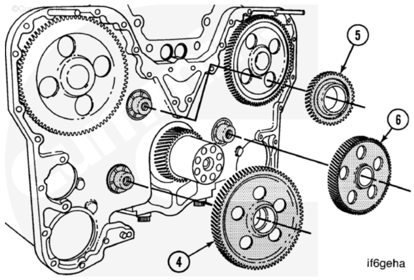
Use Lubriplate™ multi-purpose lubricant to lubricate the thrust bearings.

Install one thrust bearing (3) on each idler shaft.



⚠ CAUTION ⚠

The three idler gears, the two camshaft gears, and the crankshaft gear have index marks that must be aligned. If the marks are not aligned, the crankshaft and camshafts will not be timed properly and the engine will be extensively damaged.

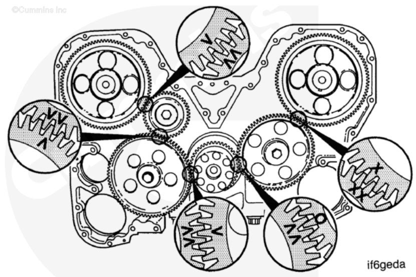


Use Lubriplate™ multi-purpose lubricant to lubricate the bushings in the gears and the shafts.

Install:

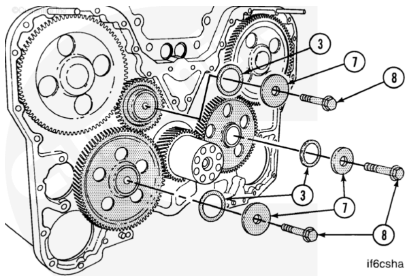
4. Water pump idler gear
5. Right bank camshaft idler gear
6. Left bank camshaft idler gear.

Align the marks that were made during disassembly.



⚠ CAUTION ⚠

The grooves on all of the thrust bearings (3) must be toward the Idler gears. If the bearings are installed wrong, failure will result

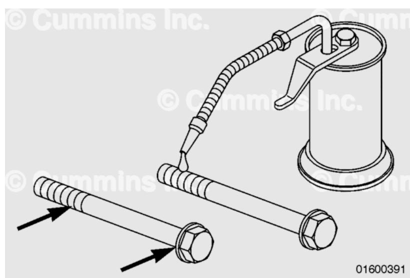


Use Lubriplate™ multi-purpose lubricant to lubricate the thrust bearings.

Install one thrust bearing (3) on each idler shaft.

Use clean engine oil. Lubricate the idler shaft capscrew threads and the flange head as shown.

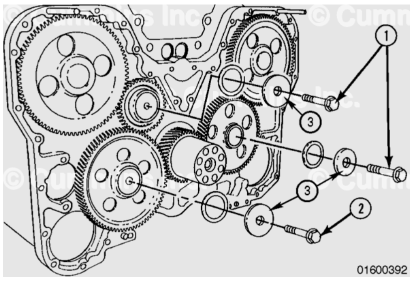
Note : The torque method for the camshaft idler and the water pump idler capscrews are different.



Install one thrust bearing retainer (3) and capscrew (1) on each camshaft idler shaft.

Capscrews with flange head:

Capscrew without flange head:	250 n.m	[185 ft-lb]



Install one thrust bearing retainer (3) and capscrew (2) onto the water pump idler shaft.

Capscrews with flanged head:

Note : In cases where the torque turn method is not possible, due to limited access, a torque value of 305 N•m [225 ft-lb] can be used in place of the torque/ turn procedure. The torque only method must only be used in extreme cases.

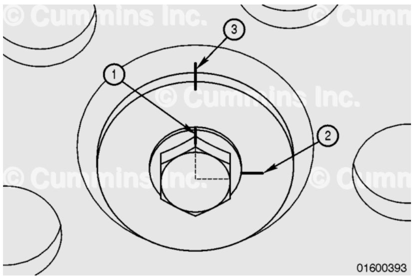
Capscrew without flange head:	250 n.m	[185 ft-lb]

To apply the 90 degree angle for the water pump idler capscrew with a flanged head, perform the following:

Once step 3 is complete (torque 88 N•m [65 ft-lb]), use a clean cloth to wipe excess oil from the thrust bearing retainer and capscrew head.

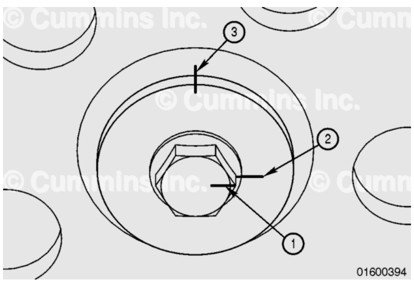
Use a marker to mark the capscrew (1) thrust bearing retainer (2), and thrust bearing retainer and idler gear (3) as indicated.

Note : Mark the capscrew at one of the points of the hexagon head (1). The 90 degrees will be located midway on the second flat in a clockwise direction from the marked point as shown in the diagram.



⚠ CAUTION ⚠

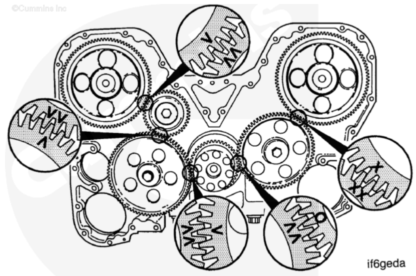
Make certain that the thrust bearing retainer done not turn while rotating the capscrew through the 90 degrees. This will cause the bolt to over torque and can result in engine failure. If the thrust bearing retainer turns, loosen the capscrews and start the torque procedure over.



Turn the capscrew clockwise through 90 degrees until the mark on the capscrew (1) meets the mark on the thrust bearing retainer (2).

Note : If a dial indicator gauge or digital angle gauge is available, use one of these to improve accuracy of the final torque angle. The use of a gauge can eliminate the need for marking the components.

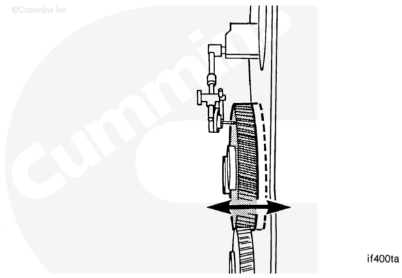
Check all of the alignment marks again to be sure that they are aligned as shown.



Use a dial indicator to measure the end clearance.

Note : The end clearance measurements are the same for the newer and older idler assemblies.

mm		in
0.13	MIN	0.005
0.46	MAX	0.018

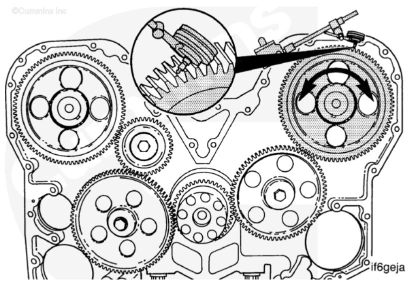


If the clearance is **not** within specifications, check for foreign material between the parts.

If the clearance is **not** within specifications, check to see if the thrust bearings are in the proper locations. Oversize thrust bearings are available.

Measure the Front Gear Train Backlash

Note : The adjacent (engaging) gear must not turn as the gear that is being measured is turned.



Use a dial indicator. Position the indicator on a gear as shown. Measure the backlash on all of the gears.

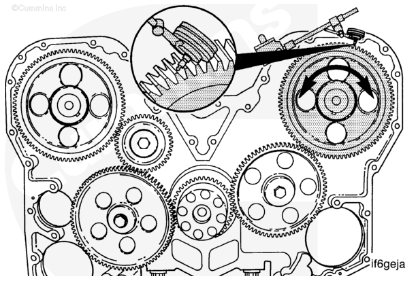
Rotate the gear being measured **clockwise**. Position the indicator at “0”. Rotate the gear **counterclockwise**. Read the indicator.

mm		in
0.07	MIN	0.003
0.51	MAX	0.020

Repeat this procedure for all of the gears except the crankshaft gear.

If the backlash is **not** with specifications, be sure the adjacent (engaging) gear is **not** moving during the measurement.

If the backlash is **not** within specifications, the gear **must** be replaced.



Finishing Steps

⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.

- Install the gear cover and all related components. Refer to Refer to Procedure 001-031 (Gear Cover, Front) in Section 1. (/qs3/pubsys2/xml/en/procedures/28/28-001-031-tr.html)
- Operate the engine and check for leaks.

Last Modified: 21-Sep-2009